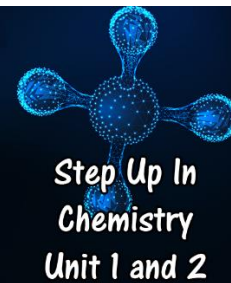


1.2 The Periodic Table

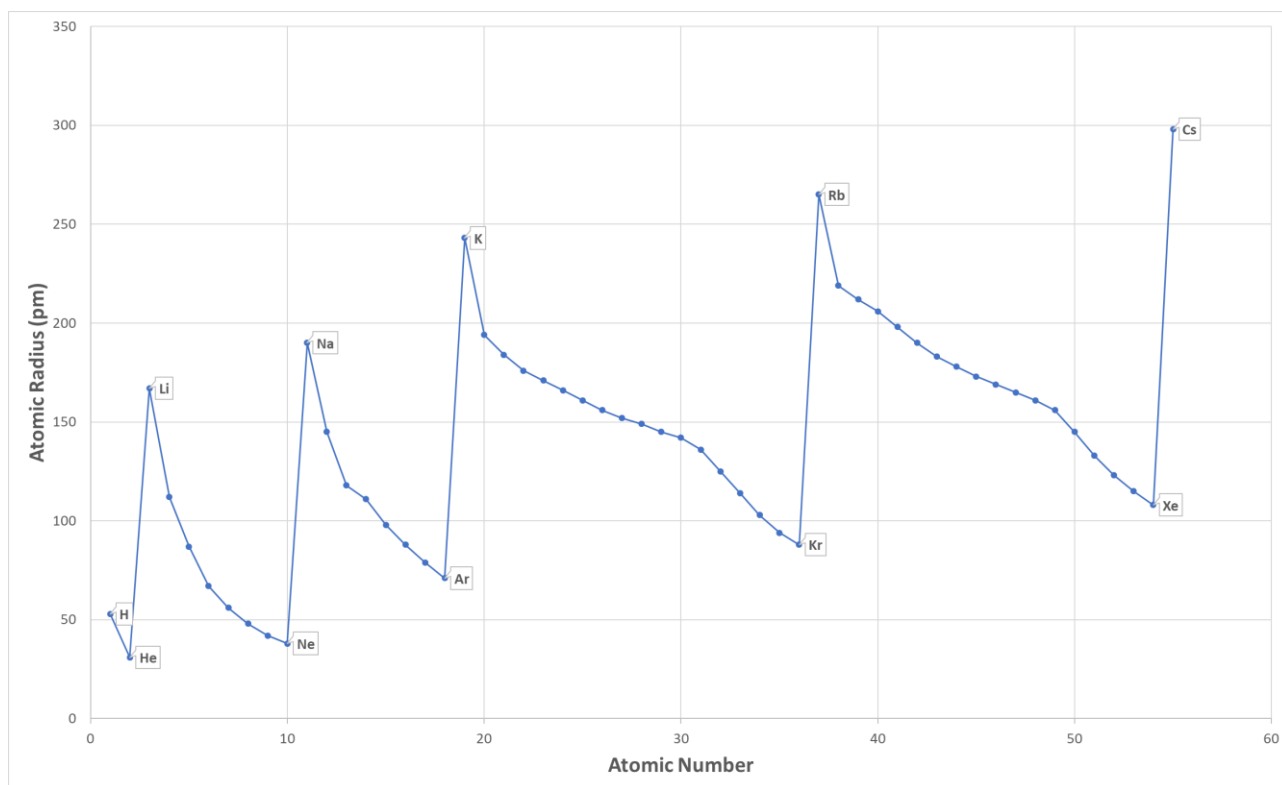
Problems Worksheet



1. The modern periodic table is credited to Dimitri Mendeleev.
 - a. What was it that made it difficult to order the elements in a table during the 19th century?
 - b. What was unique about Mendeleev's periodic table?
 - c. Which group of elements were not on Mendeleev's periodic table? Why was this so?
2. Describe the following groupings in the periodic table.
 - a. Groups
 - b. Periods

3. Write the electron configuration of the following elements. You should be able to do this without looking at a periodic table.
- a. The element in period 3, group 1.
 - b. The element in period 2, group 16.
 - c. The element in period 4, group 2.
 - d. The element in period 3, group 14.
 - e. The element in period 2, group 13.
 - f. The element in period 3, group 15.
4. The core charge of an atom follows a trend in the period table.
- a. What is meant by core charge?
 - b. State and explain the trend in core charge from left to right across the periods of the period table.
 - c. State and explain the trend in core charge from the top to the bottom of groups in the periodic table.

5. The graph below shows the trend in atomic radius for the first 55 elements in the period table.



State and explain the trend shown.

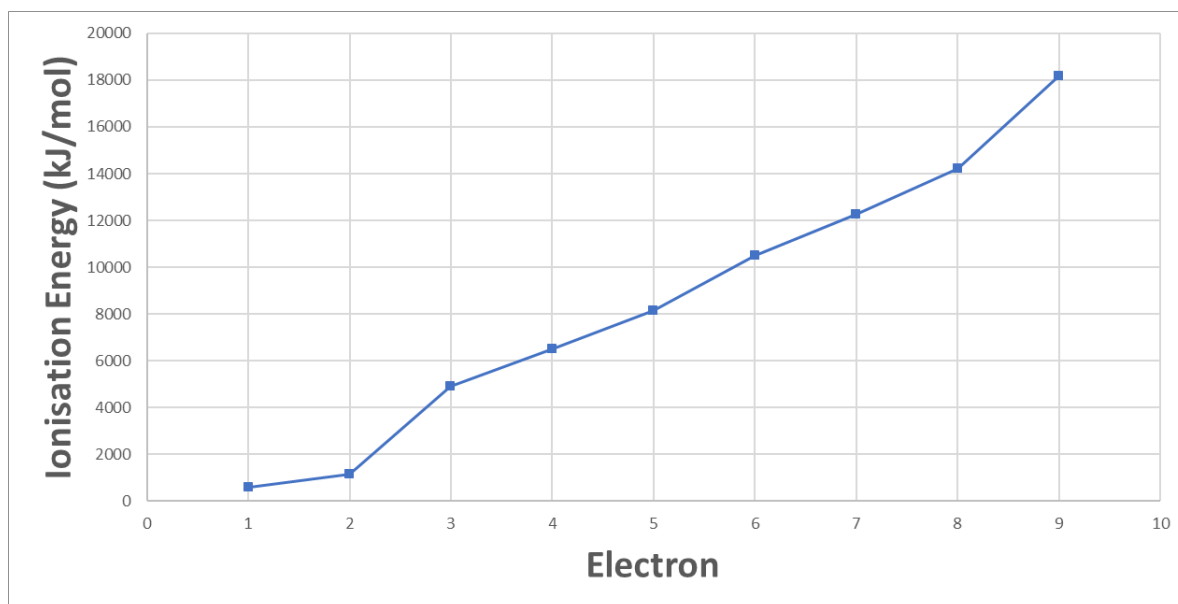
6. Define electronegativity and explain its trend in the periodic table.

7. The periodic table is shown below.

1 H hydrogen 1.008																	2 He helium 4.003						
3 Li lithium 6.94	4 Be beryllium 9.012																	5 B boron 10.82	6 C carbon 12.01	7 N nitrogen 14.01	8 O oxygen 16.00	9 F fluorine 19.00	10 Ne neon 20.18
11 Na sodium 22.99	12 Mg magnesium 24.31																	13 Al aluminium 26.98	14 Si silicon 28.09	15 P phosphorus 30.97	16 S sulfur 32.07	17 Cl chlorine 35.45	18 Ar argon 39.95
19 K potassium 39.10	20 Ca calcium 40.08	21 Sc scandium 44.96	22 Ti titanium 47.88	23 V vanadium 50.94	24 Cr chromium 52.00	25 Mn manganese 54.94	26 Fe iron 55.85	27 Co cobalt 58.93	28 Ni nickel 58.69	29 Cu copper 63.55	30 Zn zinc 65.38	31 Ga gallium 69.72	32 Ge germanium 72.59	33 As arsenic 74.92	34 Se selenium 78.96	35 Br bromine 79.90	36 Kr krypton 83.80						
37 Rb rubidium 85.47	38 Sr strontium 87.62	39 Y yttrium 88.91	40 Zr zirconium 91.22	41 Nb niobium 92.91	42 Mo molybdenum 95.94	43 Tc technetium	44 Ru ruthenium 101.1	45 Rh rhodium 102.9	46 Pd palladium 106.4	47 Ag silver 107.9	48 Cd cadmium 112.4	49 In indium 114.8	50 Sn tin 118.7	51 Sb antimony 121.8	52 Te tellurium 127.6	53 I iodine 126.9	54 Xe xenon 131.3						
55 Cs caesium 132.9	56 Ba barium 137.3	57–71 *La lanthanum 138.9	72 Hf hafnium 178.5	73 Ta tantalum 180.9	74 W tungsten 183.9	75 Re rhenium 186.2	76 Os osmium 190.2	77 Ir iridium 192.2	78 Pt platinum 195.08	79 Au gold 197.0	80 Hg mercury 200.6	81 Tl thallium 204.4	82 Pb lead 207.2	83 Bi bismuth 209.0	84 Po polonium	85 At astatine	86 Rn radon						
87 Fr francium	88 Ra radium 226.0	89–103 **Ac actinium	104 Rf rutherfordium	105 Db dubnium	106 Sg seaborgium	107 Bh bohrium	108 Hs hassium	109 Mt meitnerium	110 Ds darmstadtium	111 Rg roentgenium	112 Cn copernicium	113 Nh nihonium	114 Fl flerovium	115 Mc moscovium	116 Lv livermorium	117 Ts tennessine	118 Og oganesson						
* Lanthanide series		58 Ce cerium 140.1	59 Pr praseodymium 140.9	60 Nd neodymium 144.2	61 Pm promethium	62 Sm samarium 150.4	63 Eu europium 152.0	64 Gd gadolinium 157.3	65 Tb terbium 158.9	66 Dy dysprosium 162.5	67 Ho holmium 164.9	68 Er erbium 167.3	69 Tm thulium 168.9	70 Yb ytterbium 173.0	71 Lu lutetium 175.0								
** Actinide series		90 Th thorium 232.0	91 Pa protactinium	92 U uranium 238.0	93 Np neptunium	94 Pu plutonium	95 Am americium	96 Cm curium	97 Bk berkelium	98 Cf californium	99 Es einsteinium	100 Fm fermium	101 Md mendelevium	102 No nobelium	103 Lr lawrencium								

- Define first ionisation energy.
- Show the trends in first ionisation energy on the diagram above.
- Explain the trends shown.

8. The successive ionisation energies for an element are shown below.



- Given that the element is from the fourth period, identify the element.
- Write the electron configuration for the element.
- Explain the trend in the successive ionisation energies for the element.
- Write equations for the successive ionisations of the element to remove its valence electron(s). Use the graph to estimate the ionisation energies.